

Beyond the Scale: Modeling Anastomotic Leak as BMI Increases

12/10/25

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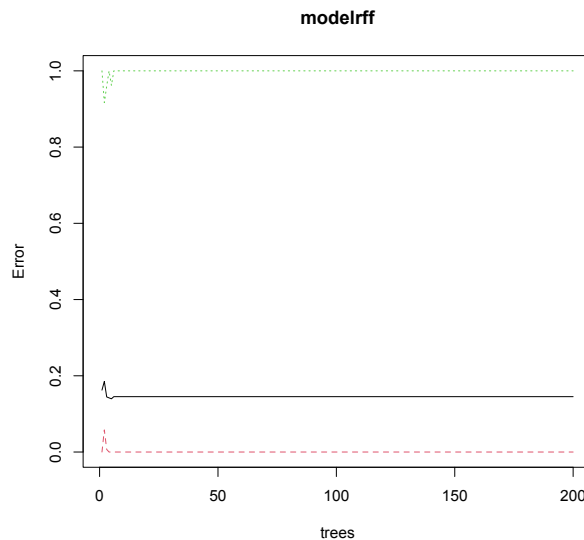


Figure 1: Random Forest Model

Introduction

This report analyzes the risks of anastomotic leaking following a colectomy at LSU. The dataset, collected by physician Meredith Grey, contains information on 179 patients, including their heights, weights, gender, tobacco usage, operative length, and if they contain the response variables for anastomotic leaking. Using the dataset, Dr. Grey seeks assistance to graphically represent the dangers of anastomotic leaking with a patient's body mass index (BMI). More specifically, a research model that (a) Describes the used to analyze the data and (b) how the model determines a BMI range sample impacts two specific cases.

Model Selection: Random Forests

To analyze the risks of anastomotic leaking following a colectomy procedure, we can build a Random Forest classification model using patient demographic and clinical predictors, with the response variable for anastomotic leaking as a binary outcome (0 = no leak, 1 = leak). The predictors include a patients BMI levels, age, gender, height, weight, race, tobacco usage, diabetes probability, coronary or pulmonary artery disease predictors, cancer probability, albumin measure, and operation length in days. All categorical predictors were converted to factors, and the response variable was also converted into a factor so that the algorithm performed classification rather than regression.

A random forest model was chosen for this dataset because it is a method that improves accuracy and reduces overfitting relative to a single decision tree. Since the medical relationships in this dataset may be nonlinear and may involve interactions between variables, tree-based methods are well suited for capturing these complexities without requiring explicit specification of interactions or transformations. A single classification tree is easy to interpret but typically has high variance and can overfit small medical datasets. Random forests handle nonlinearities, provide internal cross-validation through error estimation, and return variable importance measures that help identify which clinical factors are most associated with leak risk.

To tune the model, I set the number of trees (ntree) to 200, used mtry = 2 to control the number of randomly selected predictors at each split, and increased nodesize to prevent overly small terminal nodes,

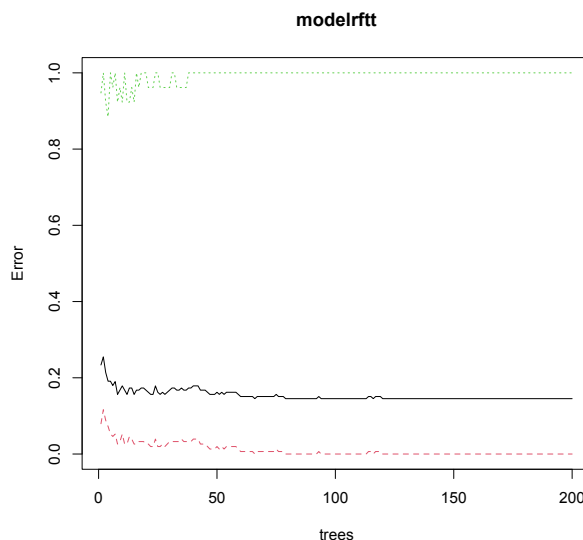


Figure 2: Tuned Random Forest Model

which helps reduce overfitting given the modest sample size. The tuned random forest model outperforms the original model because hyperparameter optimization produced a forest with lower error rates and more stable predictive performance. After tuning the number of variables tried at each split (`mtry`), the minimum node size, and the sampling size, the model achieved a substantially improved cross-validated loss and a noticeably lower error curve. This means the tuned model predicts leak probability more accurately and with less variability. Because of this stronger overall performance, the tuned model provides more reliable risk estimates for the case studies involving Arizona Robbins and Richard Webber.

The Side Effects of BMI's

Body Mass Index (BMI) is one of the key predictors in determining the possible risk of anastomotic leak after a colectomy operation. To better understand how BMI influences postoperative outcomes, we examined two case studies by holding all patient characteristics constant except BMI, which we varied from 15 to 45. This approach isolates the impact of obesity while preserving each patient's unique clinical profile. Keep in mind, a BMI level of under 18.5 is considered underweight, BMI's of 18.5-24 are considered normal, 25-35 are considered overweight, and any BMI level over 35 is considered obese.

Because the random forest model captures nonlinear relationships and interactions between risk factors, the resulting BMI-risk curves are offering insight into how BMI interacts with age, comorbidities, albumin levels, and operative time to influence the probability of an anastomotic leak.

Case 1: Arizona Robbins

To explore how BMI affects the risk of anastomotic leaking for a specific patient, we can hold all of Arizona Robbins's characteristics constant and varied only her BMI across a clinically meaningful range (15–45). Using the fitted classification model (random forest or logistic regression), we predicted her probability of leak at each BMI value. The probabilities are then graphed BMI on the x-axis and predicted leak probability on the y-axis.

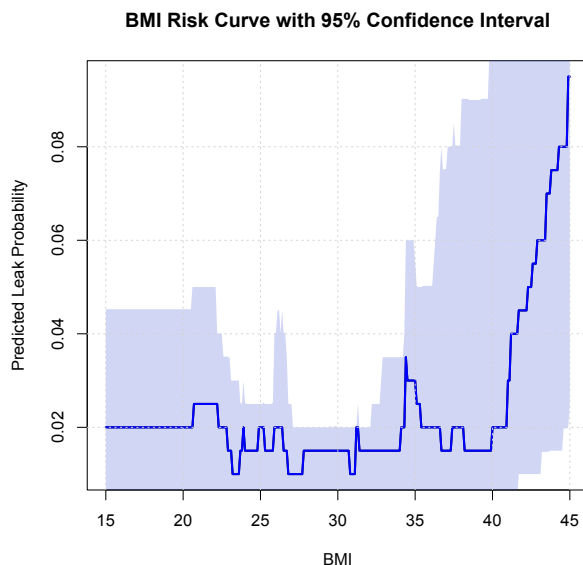


Figure 3: Predicted Leak Risk as BMI Increases for Arizona Robbins

The graph demonstrates that the effect of BMI on anastomotic leak is not uniform across patients. Arizona Robbins is a young 35-year-old female with no comorbidities, a post-operative albumin level of 4.2 and an operation length of 90 minutes (0.0625 fraction of a day). From a BMI of 15 to about 21, Arizona's probability of containing an anastomotic leaking post operation remains steady at about 2%. With her BMI of 22 to 33, her probability of a possible anastomotic leakage can range as high as less than 4% or as low as 0-1 percent before slowly increasing to a high of about 10% probability for anything above a BMI of 34. This is likely due to her clinical profile represents a very low-risk patient resulting in a better, near perfect, post operation outcome. For Arizona Robbins, who is young and has no comorbidities, BMI does not meaningfully increase risk until BMI exceeds approximately 35.

Case 2: Richard Webber

For the second case study, we can evaluate how BMI affects anastomotic leak risk for Richard Webber, a 62-year-old African American male with tobacco use, diabetes, low albumin levels of 2.8, and a long operative time of 210 minutes (0.1458 fraction of a day). Same as before, all of his characteristics were held constant while BMI was systematically varied from 15 to 45. We used the fitted classification model to estimate Richard's predicted leak probability at each BMI level and graphed the resulting risk curve.

Since Richard has multiple clinical risk factors ranging from tobacco use, diabetes, low albumin, and long surgery duration, his predicted baseline risk is substantially higher than Arizona Robbins's. Richard Webber's probability of an anastomotic leaking post operation stays at 28% when BMI levels are less than 21 before decreasing for a little while to go as low as a little over 26% for BMI levels of 21 to 24. From BMI levels of 25 and above, the probability of an anastomotic leakage starts to gradually increase reaching as high as almost 40% likely post colectomy. The resulting BMI risk curve demonstrates that obesity sharply increases the danger of anastomotic leak for patients with multiple comorbidities like Richard Webber. Compared to younger, healthier patients such as Arizona Robbins, patients with similar comorbidities as Richard Webber, have a higher chance of anastomotic leaking post colectomy operation at any BMI levels.

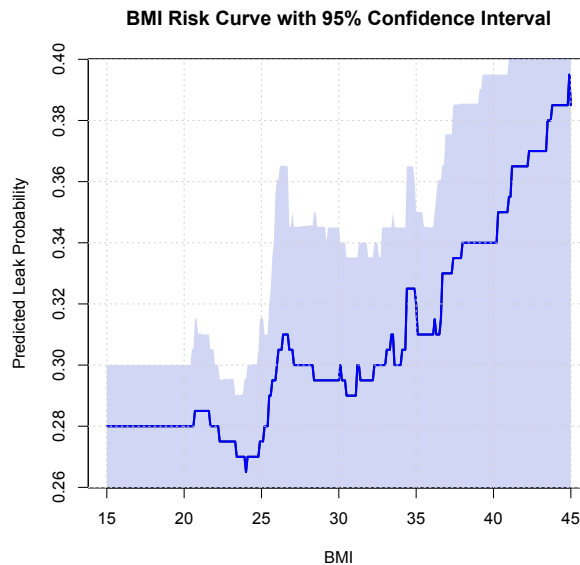


Figure 4: Predicted Leak Risk as BMI Increases for Richard Webber

Conclusion

In this analysis, we built a random forest model to better understand the potential risks of anastomotic leaking following a colectomy procedure. After cleaning the data, transforming variables, and converting categorical predictors into factors, we trained and tuned a classification model designed to capture complex nonlinear relationships among clinical variables. Tuning the model significantly improved predictive accuracy, yielding more stable and trustworthy risk estimates.

Using this model, we examined how BMI influences leak risks for two cases with two contrasting patients. The results showed that BMI does not affect all patients equally. For a young, healthy patient like Arizona Robbins, leak risk remains very low across most BMI values and only begins to rise at high levels of obesity. In contrast, for a higher-risk patient like Richard Webber who contains multiple comorbidities and a long operative time, BMI amplifies leak risk as it increases.

Appendix: R Program Code

(to perform all tests described in report)

```
## Midterm 2

# change into working directory
setwd("/Users/katelynnguyen/Downloads/439")

# read data table
data = read.csv("colon.csv",h=T)
categories = names(data)

patientidentifier = data[,1]
diagnosis = data[,2]
cptcode = data[,3]
procedure = data[,4]
gender = data[,5]
height = data[,6]
weight = data[,7]
bmi = data[,8]
age = data[,9]
race = data[,10]
tobacco = data[,11]
dm = data[,12]
cadpad = data[,13]
cancer = data[,14]
albumin = data[,15]
incisionstart = data[,16]
incisionclose = data[,17]
operativelength = data[,18]
anastomicleak = data[,19]

# part a: describe the model created to analyze the data. how did you settle on this model?
genderr = factor(gender)
racee = factor(race)
tobaccoo = factor(tobacco, levels=c(0,1))
dmm = factor(dm, levels=c(0,1))
cadpadd = factor(cadpad, levels=c(0,1))
cancerr = factor(cancer, levels=c(0,1))
anastomicleakk = factor(anastomicleak, levels=c(0,1))

analysisdata = data.frame(BMI = bmi,
                          Age = age,
                          Gender = gender,
                          Race = race,
                          Tobacco = tobacco,
                          DM = dm,
                          CADPAD = cadpad,
                          Cancer = cancer,
```

```
      Albumin = albumin,
      OperativeLength = operativelength,
      Leak = anastomicleak)

model= lm(Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD +
          Cancer + Albumin + OperativeLength,
          data = analysisdata)
summary(model)
par(mfrow = c(2,2))
plot(model)

predictors = c("BMI", "Age", "Gender", "Race", "Tobacco", "DM", "CAD_PAD", "Cancer", "Albumin", "Operat
response = "Anastomotic.Leak"

# convert predictors to factors
analysisdata$Gender = factor(analysisdata$Gender)
analysisdata$Race   = factor(analysisdata$Race)
analysisdata$Tobacco = factor(analysisdata$Tobacco)
analysisdata$DM      = factor(analysisdata$DM)
analysisdata$CADPAD  = factor(analysisdata$CADPAD)
analysisdata$Cancer  = factor(analysisdata$Cancer)

analysisdata$Leak = as.factor(analysisdata$Leak)

analysisdata$Leak = factor(analysisdata$Leak, levels = c(0,1))

library(randomForest)
modelrff = randomForest( Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD + Cancer + Albumin + 
  data = analysisdata,
  ntree = 200,
  mtry = 1,
  nodesize = 10,
  sampsize = 80
)
plot(modelrff)

RF.CV.LL.tune = function(n.trees, node.size, m.try, samp.size, seed=1){
  set.seed(seed)
  n = nrow(analysisdata2)

  fold_id = sample(rep(1:10, length.out = n))
  CV.LL = 0
  for(k in 1:10){
    test  = analysisdata2[fold_id == k, ]
    train = analysisdata2[fold_id != k, ]
    temp.model = randomForest(
      Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD + Cancer + Albumin + OperativeLength,
      data = train,
      ntree = n.trees,
```

```
mtry = m.try,
nodesize = node.size,
sampsize = samp.size
)
test.y = as.numeric(test$Leak2) - 1
test.p.hat = predict(temp.model, newdata = test, type = "prob")[,2]
test.p.hat[test.p.hat == 0] = 1e-6
test.p.hat[test.p.hat == 1] = 1 - 1e-6
CV.LL = CV.LL + sum(test.y*log(test.p.hat) + (1-test.y)*log(1-test.p.hat))
}
return(CV.LL)
}
plot(CV.LL)

# test tuned data
RF.CV.LL.tune(n.trees=200, node.size=10, m.try=1, samp.size=80, seed=37)
RF.CV.LL.tune(n.trees=200, node.size=5, m.try=2, samp.size=90, seed=37)
RF.CV.LL.tune(n.trees=500, node.size=5, m.try=3, samp.size=120, seed=37)
RF.CV.LL.tune(n.trees=1000, node.size=5, m.try=4, samp.size=150, seed=37)

modelrftt = randomForest(
  Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD + Cancer + Albumin + OperativeLength,
  data = analysisdata,
  ntree = 200,
  mtry = 2,
  nodesize = 5,
  sampsize = 90
)
plot(modelrftt)

bmirange = seq(15,45,by = 0.1)

arizonaa = data.frame( Age = 35,
  Gender = factor("Female", levels = levels(analysisdata$Gender)),
  Race = factor("White", levels = levels(analysisdata$Race)),
  Tobacco = factor("0", levels = levels(analysisdata$Tobacco)),
  DM = factor("0", levels = levels(analysisdata$DM)),
  CADPAD = factor("0", levels = levels(analysisdata$CADPAD)),
  Cancer = factor("0", levels = levels(analysisdata$Cancer)),
  Albumin = 4.2,
  OperativeLength = 90/1440 # fraction of day
)

apred1 = cbind( data.frame(BMI = bmirange), arizonaa)
apred1$LeakRisk = predict(modelrftt, newdata = apred1, type = "prob")[,2]
plot(apred1$BMI, apred1$LeakRisk,
  type = "l", lwd = 2,
  xlab = "BMI",
  ylab = "Predicted Probability of Anastomotic Leak",
```



```
main = "Predicted Leak Risk for Arizona Robbins\nas BMI Increases")

grid()

# confidence interval
set.seed(123)
pred.mat = matrix(NA, nrow = length(bmirange), ncol = 200)
for(b in 1:200){
  boot.rows = sample(1:nrow(analysisdata), replace = TRUE)
  boot.rf = randomForest( Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD + Cancer +
    Albumin + OperativeLength,
    data = analysisdata[boot.rows, ],
    ntree = 200,
    mtry = 2,
    nodesize = 5,
    sampsize = 90
  )
  pred.mat[, b] = predict(boot.rf, newdata = apred1, type = "prob")[,2]
}
lower.CI = apply(pred.mat, 1, quantile, probs = 0.05)
upper.CI = apply(pred.mat, 1, quantile, probs = 0.95)

plot(apred1$BMI, apred1$LeakRisk,
  type="l", lwd=2,
  xlab="BMI", ylab="Predicted Leak Probability",
  main="BMI Risk Curve with 95% Confidence Interval")

# Add CI as shaded region
polygon(c(bmirange, rev(bmirange)),
  c(lower.CI, rev(upper.CI)),
  col=rgb(0.1, 0.2, 0.8, 0.2), border=NA)

# Add point estimate line
lines(bmirange, point.pred, lwd=2, col="blue")
grid()

richardd = data.frame( Age = 62,
  Gender = factor("Male", levels = levels(analysisdata$Gender)),
  Race = factor("AA", levels = levels(analysisdata$Race)),
  Tobacco = factor(1, levels = levels(analysisdata$Tobacco)),
  DM = factor(1, levels = levels(analysisdata$DM)),
  CADPAD = factor(0, levels = levels(analysisdata$CADPAD)),
  Cancer = factor(0, levels = levels(analysisdata$Cancer)),
  Albumin = 2.8,
  OperativeLength = 210/1440 # convert into fraction of day
)

rpred1 = cbind( data.frame(BMI = bmirange), richardd)
rpred1$LeakRisk = predict(modelrftt, newdata = rpred1, type = "prob")[,2]
```

```
plot(rpred1$BMI, rpred1$LeakRisk,
     type = "l", lwd = 2,
     xlab = "BMI",
     ylab = "Predicted Probability of Anastomotic Leak",
     main = "Predicted Leak Risk for Richard Webber \nas BMI Increases")

grid()

# confidence interval for RW
set.seed(123)
pred.mat = matrix(NA, nrow = length(bmirange), ncol = 200)
for(b in 1:200){
  boot.rows = sample(1:nrow(analysisdata), replace = TRUE)
  boot.rf = randomForest( Leak ~ BMI + Age + Gender + Race + Tobacco + DM + CADPAD + Cancer +
    Albumin + OperativeLength,
    data = analysisdata[boot.rows, ],
    ntree = 200,
    mtry = 2,
    nodesize = 5,
    sampsize = 90
  )
  pred.mat[, b] = predict(boot.rf, newdata = rpred1, type = "prob")[,2]
}
lower.CI = apply(pred.mat, 1, quantile, probs = 0.05)
upper.CI = apply(pred.mat, 1, quantile, probs = 0.95)

plot(apred1$BMI, rpred1$LeakRisk,
     type="l", lwd=2,
     xlab="BMI", ylab="Predicted Leak Probability",
     main="BMI Risk Curve with 95% Confidence Interval")

# Add CI as shaded region
polygon(c(bmirange, rev(bmirange)),
       c(lower.CI, rev(upper.CI)),
       col=rgb(0.1, 0.2, 0.8, 0.2), border=NA)

# Add point estimate line
lines(bmirange, rpred1$LeakRisk, lwd=2, col="blue")
grid()
```